

Worksheet 9 *Nullclines and two-species equilibria*

Prepares you for lab manual pages **GLV1, GLV2, GLV3, GLV4** (section L3 S5)

Name:

Name of TA:

TO DO: Worksheet 9 is on the Canvas schedule but did not exist in 2025. It should give mechanical practice for GLV1–GLV4: writing a two-species change equation from a description (moose and squirrel; gerbils), drawing nullclines, and locating equilibria in state space. Draft below; replace freely.

1. Two species P and Q obey $P' = P(1 - P/200) - 0.2PQ$ and $Q' = 0.5Q(1 - Q/100) - 0.1PQ$.

- (a) Is each species helped or harmed by the other? Refer to the interaction terms.
- (b) Find the P -nullcline (where $P' = 0$) and the Q -nullcline (where $Q' = 0$), and draw them in the state space below.
- (c) Mark every equilibrium point with a dot.

